

University of Hertfordshire

7WCM2034 Fundamentals of Software Engineering

Coursework

60 Marks

You have to answer all the below questions. Each question has different marks based on their difficulties. Exercise solutions must be uploaded to Canvas in PDF format!

1. Why is software engineering so important in today's world? Articulated succinctly in a few sentences. **(3 Marks)**.
2. Explore European ICT Professional Profiles (look in course materials or [download from this link](#) and fill in the following blanks **(5 Marks)**.
 - a. _____ is accountable for the implementation and integration of software or systems.
 - b. SLAs, OLAs, and KPIs requirements are defined by _____.
 - c. Quality policy established and deployed by _____.
 - d. _____ perform effective training in the classroom, online or informally.
 - e. Risks, threats, and consequences are evaluated by _____.
 - f. _____ define and implement procedures linked to IS security.
 - g. Staff coordinator and manager is _____.
 - h. _____ develop tools to increase test effectiveness.
3. Explore the Software Engineering Body of Knowledge (SWEBOK V3) which was developed by the Software Engineering Coordinating Committee and contains recommended practices, needed knowledge, ethical and professional standards (look in the course materials or download from this [link](#)) and answer the following questions: **(5 Marks)**
 - a. Briefly define Software Requirements.
 - b. What is the difference between functional and non-functional requirements?
 - c. Outline the Software Design Principles.
 - d. Describe the nature of software maintenance.
4. Try to create a "Big Picture" of software development. Write down SDLC phases and then divide them into European ICT Professional Profiles and Software Engineering sub-disciplines (from the first lecture) in the following phases **(5 Marks)**.
 - a. Requirement gathering and analysis
Sub-disciplines: Requirements engineering, ...
 - b. Design
 - c. Implementation
Professionals: Software developer, ...
 - d. Testing
 - e. Deployment and maintenance

More questions on the next page.

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5. Select the professional profile by your last student ID Number (example B6011X - where X is your last student ID Number) prepare a little verbal presentation and describe in your own words about profession and roles on different SDLC stages (**3 Marks**).

0. ICT SECURITY SPECIALIST
1. DEVELOPER
2. PROJECT MANAGER
3. QUALITY ASSURANCE MANAGER
4. SERVICE DESK AGENT
5. SYSTEMS ANALYST
6. SYSTEMS ARCHITECT
7. TEST SPECIALIST
8. SYSTEMS ADMINISTRATOR
9. SERVICE DESK AGENT

6. Read all 12 Principles that are behind the Agile Manifesto, available at this [link](#). Select one that corresponds to your last student ID Number (example B6011X - where X is your last student ID Number) and try to answer the following questions: (**3 Marks**)

- a. What does this mean?
- b. Why is it important?
- c. Is there some principle with the same meaning?

More questions on the next page.

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For the next question, you need to first explore some CASE tools for UML.

Exercise 1 - “Learn some useful CASE tools: UML.”

- ➔ Explore at [Use Case Diagrams](#) and [UML](#)
- ➔ Explore [IBM's introduction to UML](#)
- ➔ Select and learn to use some modeling tools:
 - [draw.io](#) (Cloud-based)
 - [Modelio](#)
[YouTube Tutorial](#)
 - [Dia](#)
[YouTube Tutorial](#)

7. Create a use case diagram (max 5 actions). Select the diagram by your last student ID Number (example B6011X - where X is your last student ID Number): **(6 Marks)**

0. Using of Cloud Storage
1. Using of blog
2. Using of social network
3. Using of e-shop
4. Using of Moodle
5. Using of Search Engine
6. Using of Canvas
7. Using of e-mail
8. Using of chat
9. Using of events calendar

N.B: Export/Download the wireframe as a PNG, JPG, PDF or take screenshots and use them in the final submission coursework document according to the question number.

More questions on the next page.

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For the next question, you need to first explore some tools for creating a wireframe.

Exercise 2 - "Let's create some wireframe."

→ User interface design and 10 rules of thumb.

→ Read about Wireframing for beginners

→ Select and learn the tool:

- wireframe.cc (Cloud-based)
YouTube Tutorial
- Pencil (Desktop app)
YouTube Tutorial

8. Create a wireframe for (Select the task by your last student ID Number (example B6011X - where X is your last student ID Number): **(6 Marks)**

0. Desktop mail client.
1. Web page contact page.
2. E-shop cart page.
3. Web page blog page.
4. E-shop product details page.
5. Smartphone app registration form.
6. Instant messaging app for desktop.
7. Browser app for smartphone.
8. Web chat.
9. Scientific calculator app for desktop.

N.B: Export/Download the wireframe as a PNG, JPG, PDF or take screenshot and use them in final submission coursework document according to the question number.

More questions on the next page.

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For the next question, you need to first explore some tools for mastering the class diagram.

Exercise 3 - “Mastering the class diagram”

➔ Explore [Class Diagrams](#) and [Class Diagram Guidelines](#)

➔ Select and learn the tool:

The recommended tool is [Visual Studio 2017](#), but if you want, you can use any other tools, for example:

- [draw.io](#) (Cloud-based)
- [Modelio](#)
[YouTube Tutorial](#)

9. Create a simple (one) Class diagrams for following object (Select the diagram by your last student ID Number (example B6011X - where X is your last student ID Number) and object must have minimum 3 attributes with different visibility and minimum 3 operations: **(6 Marks)**

0. Doctor
1. House
2. Cat
3. User account
4. Window
5. Shop
6. Oven
7. Car
8. Door
9. Manager

N.B: Export/Download the wireframe as a PNG, JPG, PDF or take screenshot and use them in the final submission coursework document according to the question number.

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For the next question, you need to first explore some software framework resources.

Exercise 4 - “Meet software framework”

→ [Understanding software frameworks](#)

→ PHP framework named [CodeIgniter](#)

10. Describe in few sentences in your own word and explanation: **(3 Marks)**

- a. What architectural design patterns does it implement?
- b. Why is it helpful in development?

For the next question, you need to first explore how to use a documentation generator and the comment Java class below using “special comment blocks”.

Exercise 5 – “Learn how to use documentation generator and comment Java class below using “special comment blocks”.

→ Read about [Doxygen](#)

→ Read about [documenting code with it](#)

More questions on the next page.

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11. Comment on all elements (class and method descriptions, params and returns) of class below: (4 Marks)

```
package default;

public class MyClass {
    public int add(int a, int b){
        return a + b;
    }
    public String add(String a, String b){
        return a + b;
    }
    public int sub(int a, int b){
        return a - b;
    }
    public int mult(int a, int b){
        return a * b;
    }
    public int div(int a, int b){
        return a / b;
    }
    public int mod(int a, int b){
        return a%b;
    }
    public String sayHello(String name){
        return "Hello" + name;
    }
}
```

More questions on the next page.

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12. Design an algorithm using a flowchart. Select the algorithm by your last student ID Number (example B6011X - where X is your last student ID Number **(5 Marks)**)

- a. Repeat the flowchart symbols <https://en.wikipedia.org/wiki/Flowchart>
- b. Go to the draw.io and select "Create New Diagram".
- c. From left pane select "Flowchart" Select the algorithm to describe by last number of student ID:

0. Go to sauna
1. Clean window
2. While cycle in while cycle
3. Drink water from river
4. Open two locked door
5. Find friends.
6. Get a job.
7. Make a phone call.
8. Buy milk.
9. Make a snowman.

For the next question, you need to first Learn to use Git version control.

Exercise 6 - "Learn to use Git version control"

1. Download and install Git
2. Explore resources for getting started
3. Have a go! Try git.
4. Go to the Sign In page and Register.
5. Create new project (public repository).
6. Push or create file that contains your name or student number into created repository.
7. Push files from exercises one and two to created git repository.

13. Based on the above exercises, put a link to the repository with exercise solutions as based on the given step-by-step instruction **(6 Marks)**.

End of questions.